THADOMAL SHAHANI ENGINEERING COLLEGE

DEPARTMENT OF

Artificial Intelligence & Data Science

Name of Student : Adit Sanjay Mokashi

Class : SE

Division : S2

Roll Number : 2511136

Batch : S23

Subject : Fundamentals of Telecommunication Systems

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Experiment Title : Demonstration of Router Hardware and Connecting Different LANs

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Experiment No. 07: Demonstration of Router Hardware and Connecting Different LANs

Aim

To identify the hardware interfaces of a Router and to design, configure, and verify the connectivity between two distinct Local Area Networks (LANs) using a Router in Cisco Packet Tracer.

Theory

1. What is a Network Router?

A **Network Router** is a networking device that connects **two or more different networks** and forwards data packets between them. It decides the **best path** for data to travel from the source network to the destination network.

Routers are commonly used to connect a **Local Area Network (LAN)** to the **Internet**.

Key Features:

- Works at the **Network Layer (Layer 3)** of the OSI model.
- Uses **IP addresses** to identify devices.
- Determines the **best path** for data transmission.
- Can provide **security features** like firewalls.
- Often includes **Wi-Fi functionality** in home networks.

Example:

When you open a website, your router forwards the request from your home network to the internet and returns the webpage data back to your device.

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2. Explain the role of a "Default Gateway" in a network

A **Default Gateway** is the **IP address of the router** that a device uses to communicate with devices **outside its own network**.

When a device wants to send data:

- If the destination is **inside the same network**, data is sent directly.

- If the destination is **outside the network**, the device sends the data to the **default gateway (router)**.

  The router then forwards the data toward the destination network.

  Example:

  If your laptop (IP: 192.168.1.5) wants to access Google:

- Google is not in the local network.
- The laptop sends the data to the default gateway (for example 192.168.1.1).
- The router forwards the data to the internet.

  Without a default gateway, devices cannot communicate outside their local network.

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3. How does a Router differ from a Switch?

Feature	Router	Switch
Function	Connects different networks	Connects devices within same network
OSI Layer	Layer 3 (Network Layer)	Layer 2 (Data Link Layer)
Address used	IP Address	MAC Address
Purpose	Routes data between networks	Transfers data within LAN
Example	Connects home network to Internet	Connects computers in office

Simple Difference:

- **Router → connects networks**
- **Switch → connects devices inside a network**

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4. What is a Routing Table and how is it used to forward packets?

A **Routing Table** is a data table stored in a router that contains information about:

- Different network destinations
- Possible paths to reach them
- The best route to send data

The router checks the routing table whenever it receives a packet.

Routing Table contains:

- Destination IP address
- Next hop (where to send packet next)
- Interface to use
- Metric (cost of route)

How it works:

1. A packet arrives at the router.
2. Router reads the **destination IP address**.
3. Router searches the routing table for the best path.
4. Packet is forwarded through the appropriate interface.

Hardware Components Required

- **PC-PT (x6):** End devices acting as network nodes (3 for LAN 1, 3 for LAN 2).
- **Switch 2960 (x2):** Two 24-port switches to create independent local networks. One switch for each LAN.
- **Router ISR 4321 (x1):** The device to connect the two LANs.
- **Cabling:** Copper Straight-Through cables for all connections.

Hardware and Connection Setup

- LAN 1 (Network ID): 192.168.1.0 | Subnet Mask: 255.255.255.0
- LAN 2 (Network ID): 10.0.0.0 | Subnet Mask: 255.0.0.0
- PC Naming Convention:
  - LAN 1: S11_01_A, S11_01_B, S11_01_C (IPs: .2, .3, .4)
  - LAN 2: S11_01_D, S11_01_E, S11_01_F (IPs: .2, .3, .4)

Procedure

Step 1: Physical Topology Design

- Place the ISR 4321 Router in the center of the workspace.

- Place two 2960 Switches (one for each LAN) and six PCs (three per switch).
- Naming: Rename PCs following the convention (e.g., S11_01_A through S11_01_F).

Step 2: Hardware Identification

- Click on the Router and select the Physical tab.
- Observe the modular slots and the GigabitEthernet ports.
- Note the power switch; ensure the router is turned ON.

Step 3: Physical Cabling

- Connect Switch 0 to Router interface GigabitEthernet 0/0/0.
- Connect Switch 1 to Router interface GigabitEthernet 0/0/1.
- Connect all PCs to their respective switches.
- Observation: Note that the link lights between the switches and the router are Red. This is because router ports are disabled by default.

Step 4: Router Interface Configuration (The "Bridge" Setup)

1. Click **Router > CLI**. Type ‘no’ at the initial setup prompt.
2. Enter the following commands to configure the gateways:
   - `Router> enable`
   - `Router# config t`
   - `Router(config)# interface gigabitEthernet 0/0/0`
   - `Router(config-if)# ip address 192.168.1.1 255.255.255.0`
   - `Router(config-if)# no shutdown`
   - `Router(config-if)# interface gigabitEthernet 0/0/1`
   - `Router(config-if)# ip address 10.0.0.1 255.0.0.0`
   - `Router(config-if)# no shutdown`

Observation: The link lights should now turn Green.

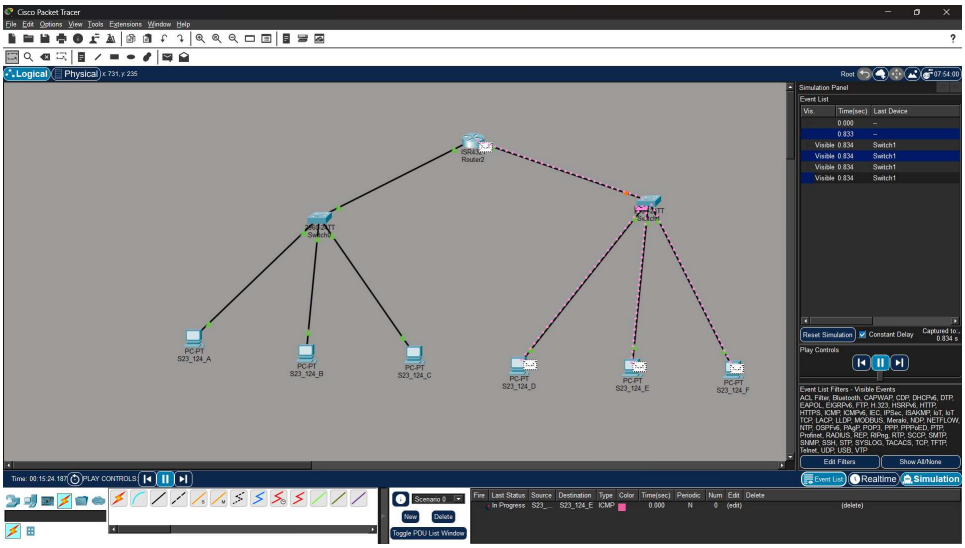
Command	Purpose / Function
enable	Moves the user from **User EXEC Mode** (view only) to **Privileged EXEC Mode**. This allows the user to view detailed configurations and enter setup modes.
config t (configure terminal)	Moves the router into **Global Configuration Mode**. This is the only mode where you can make changes that affect the router as a whole.
interface gigabitEthernet 0/0/0	Moves the router into **Interface Configuration Mode**. It tells the router, "The next commands I type are only for this specific physical port."
ip address [IP] [Mask]	Assigns a logical Identity (IP) and defines the network size (Subnet Mask) for that interface. This becomes the **Default Gateway** for that LAN.
no shutdown	By default, Cisco router ports are "Administratively Down" (turned off for security). This command **powers on** the interface to allow data flow.

Step 5: IP Addressing (Logical Setup)

- **For LAN 1 PCs:** Set IPs 192.168.1.x and set Default Gateway to 192.168.1.1.
- **For LAN 2 PCs:** Set IPs 10.0.0.x and set Default Gateway to 10.0.0.1.

Step 6: Testing and Simulation

1. **Real-time:**
   - Open Command Prompt on **S11_01_A** and type `ping 10.0.0.4`.
   - Verify connectivity. (Note: The first ping may timeout due to ARP).
2. **Simulation Mode:**
   - Filter for ICMP. Use the Add Simple PDU (Envelope) to click S11_01_A (LAN 1) then S11_01_F (LAN 2).
   - Observe the packet crossing from the 192.x network to the 10.x network through the router.



Observations

- **Gateway Necessity:** If the "Default Gateway" is not configured on the PCs, the packet never reaches the router and communication fails.
- **Routing Decision:** The Router stops the broadcast from LAN 1 and intelligently forwards the packet specifically to the interface connected to LAN 2.
- **Different Subnets:** Communication was successfully established between two completely different IPs, which is impossible with only a switch.

Conclusion

The experiment successfully demonstrates that a **Router** is the essential hardware for Inter-LAN or Inter-Network communication. It proves that while switches manage local traffic, routers are required to cross network boundaries by using logical IP addresses.